



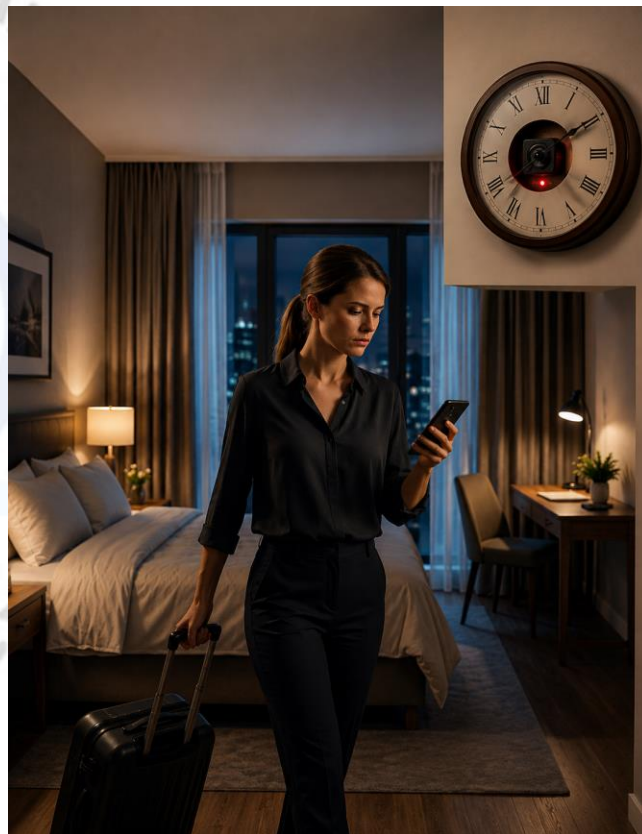
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Introduction & Problem Statement

- Hidden cameras are increasingly being misused in hotels, hostels, changing rooms, rental accommodations, and other private spaces.
- Most hidden surveillance devices are compact, wireless, and difficult to identify using conventional detection methods.
- Existing solutions are often expensive, require specialized equipment, and lack intelligent analysis capabilities for identifying suspicious devices.
- The rapid growth of IoT-enabled surveillance devices has increased privacy and safety concerns, particularly for women, travelers, and individuals in unfamiliar environments.
- There is a need for an affordable, portable, and AI-assisted detection system capable of identifying potential hidden cameras through wireless signal analysis and optical reflection techniques.





Literature Review

Research Paper (2021+)	Technique Used	Limitation
DeepDeSpy: Deep Learning-Based Wireless Spy Camera Detection System (2021)	CSI + WiFi Traffic + CNN-BiLSTM	Requires CSI extraction and deep learning processing (ResearchGate)
Detection and Localization of Hidden Wi-Fi Cameras (2022)	RSSI, CSI and localization methods	Focused mainly on localization after detection (Griffith University Research Repository)
Are There Wireless Hidden Cameras Spying on Me? (2022)	Encrypted WiFi traffic analysis	Mainly targets WiFi-based cameras only (RTCL - University of Michigan)
LocCams: Hidden Camera Detection and Localization (2023)	Wireless traffic stimulation and localization	Dependent on wireless camera behavior patterns (Semantic Scholar)
CamLoPA: Hidden Wireless Camera Localization Framework (2024)	Signal propagation path analysis	Requires traffic analysis and localization framework (arXiv)

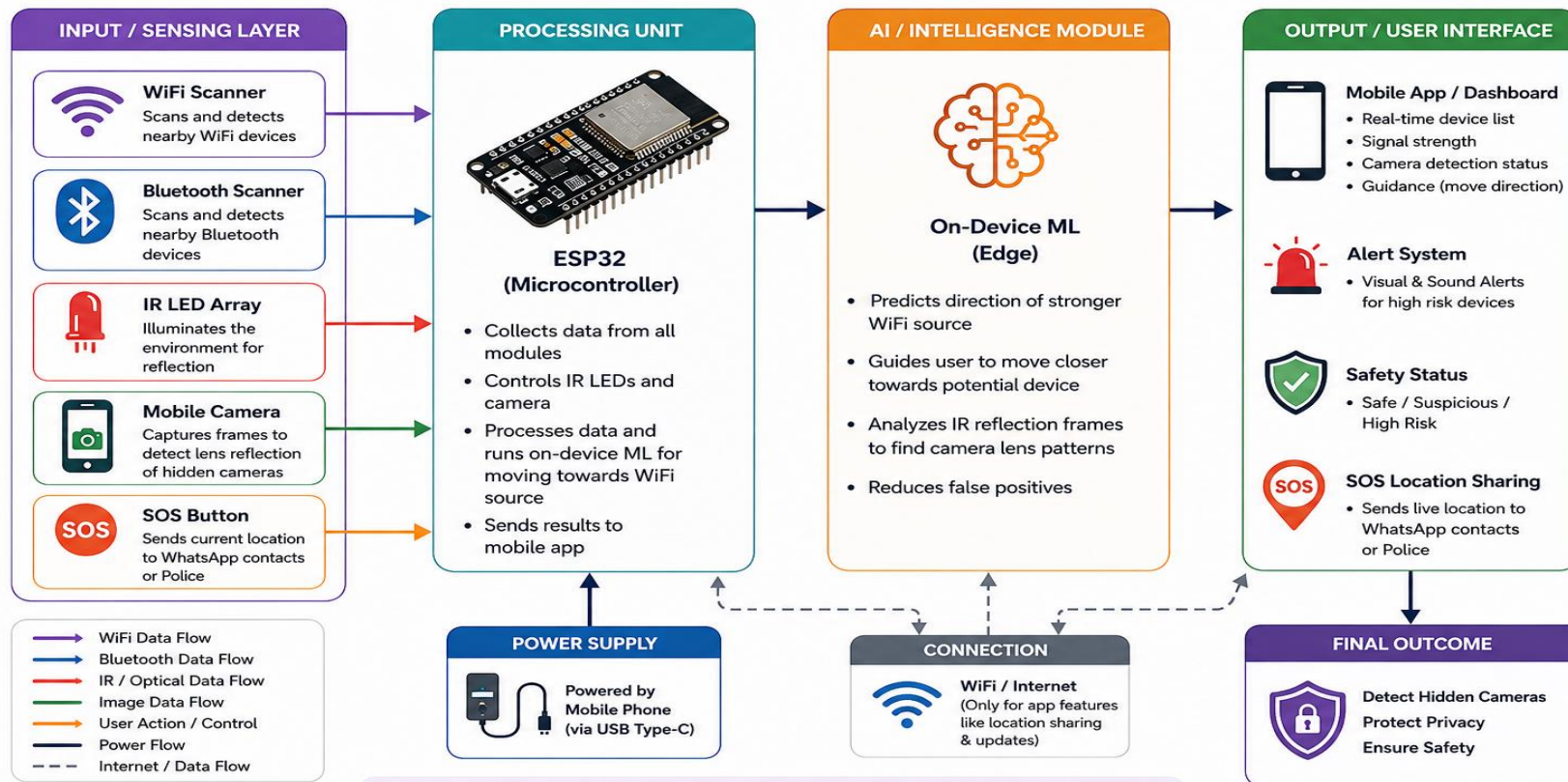


Objectives

To create an affordable, accessible, and intelligent solution for detecting potential hidden surveillance devices and protecting user privacy.

- Develop a low-cost and portable hidden camera detection system using ESP32.
- Detect suspicious wireless devices through WiFi and Bluetooth signal analysis.
- Identify potential camera lenses using IR LED reflection techniques.
- Provide real-time monitoring and device tracking through a user-friendly dashboard.
- Integrate AI-assisted analysis to improve detection accuracy and reduce false positives.
- Enhance user privacy and safety in hotels, hostels, rental accommodations, and other private spaces.

SheSafe – System Architecture



Working: ESP32 scans WiFi & Bluetooth devices, IR LEDs + Mobile Camera detect lens reflection. On-device ML guides user towards WiFi source. SOS button shares live location to WhatsApp or Police.

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Live Demonstration





Future Scope

- Replace IR LEDs with IR Laser technology for improved hidden camera lens detection accuracy.
- Integrate a photodiode sensor along with AI-based analysis for two-step verification and reduced false positives.
- Incorporate dedicated RF detection hardware to improve identification of wireless surveillance devices.
- Develop advanced AI models for intelligent threat classification and automated decision-making.
- Convert the current web dashboard into a dedicated cross-platform mobile application.

Conclusion

SheSafe is a low-cost and portable hidden camera detection system that combines WiFi scanning, Bluetooth detection, IR reflection analysis, and emergency SOS functionality to enhance user privacy and safety. By leveraging the ESP32 platform and a user-friendly dashboard, the system provides an accessible solution for detecting potential surveillance threats in hotels, hostels, rental accommodations, and other private spaces. The proposed approach demonstrates the potential of affordable IoT-based security solutions in addressing growing privacy concerns and lays the foundation for future AI-assisted detection technologies.